

Ironhack AI Engineering

COURSE SYLLABUS · 8 WEEKS

Everything below was taught live and recorded. The AI Learning Copilot searches all of it — transcripts and course notebooks together — and answers with the lesson, the notebook cell, and the recording cued to the second the topic was explained.

The eight weeks

Week	Theme	Days	Sessions	Hours
1	Python & data	5	28	20
2	Machine learning	5	19	16
3	Deep learning & vision	4	14	10
4	NLP & embeddings	4	12	8
5	Databases & LLM APIs	5	17	14
6	Deployment	2	8	6
7	LangChain & RAG	5	18	12
8	Multimodal & evaluation	2	4	4

What the copilot searches

32 lesson days	120 recordings	90 hours of teaching	64 notebooks (40 + 24 extra)	6,037 indexed passages
-------------------	-------------------	-------------------------	---------------------------------	---------------------------

How to use it

Ask in your own language	Answers come back in the language you asked in — the material stays English underneath. Spanish questions match English lessons.
Every answer names its source	Lesson day, notebook cell, and the recording cued to the second. If the copilot says a topic was not covered, that is a real answer, not a failure — nothing was invented to fill the gap.
Quiz yourself before an assessment	Pick a week or a single day and the questions are written from that material only.

Where to find it

Ask the copilot — [ai-learning-copilot-ironhack-final-project.streamlit.app](#)

Source code — [github.com/AI-Learning-Copilot/ironhack-final-project](#)

Built by — [Casilda Gil de Santivañes Finat](#) · [Felipe Martignon](#)



Scan to open

Week by week

Every session below is indexed and searchable in the copilot.

Week 1 · Python & data

D1	Kickoff — program intro, first steps in Python, and setting up Git for lab work.	239 min
D2	Python functions in depth — parameters, return values, scope, lambda functions, and list comprehensions.	184 min
D3	Using LLMs to learn effectively, then error handling and functional tools — map, filter, and reduce.	196 min
D4	Foundations for ML — tensors and linear algebra, NumPy, Pandas, and descriptive statistics.	309 min
D5	Grouping and merging data in Pandas, AI coding assistants, data visualization, Agile/Scrum basics, and object-oriented programming.	300 min

Week 2 · Machine learning

D1	Introduction to machine learning, with live demos of linear and logistic regression.	245 min
D2	Evaluating models — regression and classification metrics — and an overview of ML algorithms.	223 min
D3	A full supervised learning workflow — from problem definition and EDA through feature engineering to final evaluation.	323 min
D4	Unsupervised learning — clustering and dimensionality reduction with PCA.	112 min
D5	Kickoff for the IronKaggle competition.	30 min

Week 3 · Deep learning & vision

D1	Deep learning foundations — neurons, activation functions, backpropagation, and a first Keras model on MNIST.	202 min
D2	Computer vision — image preprocessing and convolutional neural networks.	183 min
D3	Kickoff for Project 1.	71 min
D4	Deploying models with Gradio, and improving them with data augmentation and transfer learning.	122 min

Week 4 · NLP & embeddings

D1	Regular expressions and pseudocode.	146 min
D2	NLP fundamentals — tokenization, stemming, TF-IDF, and a news-clustering example.	179 min
D3	Word embeddings, including Word2Vec.	125 min
D4	Kickoff for Project 2.	54 min

Week 5 · Databases & LLM APIs

D1	Databases — SQLite setup, CRUD operations, and table relationships.	137 min
D2	Transformer architecture, APIs, and prompt engineering.	160 min
D3	Using the OpenAI API, prompt injection, and tool calling.	184 min
D4	Connecting to AI models, quantization, and the geopolitics of AI.	145 min
D5	Mini-project demos, plus transfer learning with LoRA and QLoRA.	184 min

Week 6 · Deployment

D1	Building and deploying a full-stack AI app — a FastAPI backend on Render and a React frontend on Vercel.	269 min
D5	Group project presentations.	93 min

Week 7 · LangChain & RAG

D1	Introduction to LangChain and chains.	139 min
D2	Retrieval-augmented generation — indexing and retrieval.	120 min
D3	LangChain memory, RAG pipelines, and retrieval agents.	159 min
D4	Streaming responses, tool calling, and open models.	161 min
D5	Week recap and the IronStore mini-project.	152 min

Week 8 · Multimodal & evaluation

D1	Multimodal search with CLIP and multimodal RAG.	113 min
D2	NLP evaluation metrics.	129 min